

- 2 (a) e.g. moving in random (rapid) motion of molecules/atoms/particles
 no intermolecular forces of attraction/repulsion
 volume of molecules/atoms/particles negligible compared to volume of container
 time of collision negligible to time between collisions
(1 each, max 2) B2 [2]
- (b) (i) 1. number of (gas) molecules B1 [1]
 2. mean square speed/velocity (of gas molecules) B1 [1]
- (ii) either $pV = NkT$ or $pV = nRT$ and links n and k
 and $\langle E_k \rangle = \frac{1}{2}m\langle c^2 \rangle$ M1
 clear algebra leading to $\langle E_k \rangle = \frac{3}{2}kT$ A1 [2]
- (c) (i) sum of potential energy and kinetic energy of molecules/atoms/particles M1
 reference to random (distribution) A1 [2]
- (ii) no intermolecular forces so no potential energy B1
 (change in) internal energy is (change in) kinetic energy and this is
 proportional to (change in) T B1 [2]

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2 (a) (i) amplitude remains constant

P1 [1]